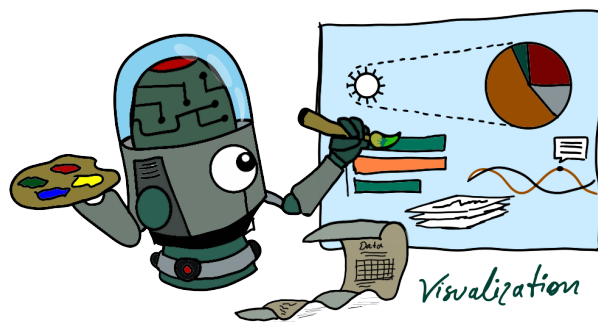


DS5343 - Visualización de Datos



Universidad de Ingeniería y Tecnología



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Graded Activity: Multidimensional Data Visualization

Goal: Apply multidimensional visualization techniques for data analysis. Use visualization techniques for high-dimensional (HD), and low-dimensional techniques.

Data:

In this activity, you will work with data on SPOTIFY data set [Kaggle](#). Feel free to enrich your dataset with additional information of Artist, Genres, and Year (contained in the same link).

Assignment

In this assignment, you will design and implement a **web application** (using Flask, Dash, Streamlit, or any other suitable framework) to analyze the dataset using multidimensional visualization techniques.

- You must define at least **three analytical tasks** that their visualizations will answer (ie. How do Spotify tracks cluster by audio features (energy, valence, danceability)?).
- it is mandatory to use at least one visualization of each of the following:
 - **RadViz** (dimensional anchoring).
 - **Star Coordinates** (with axis weighting or dragging if possible).
 - **Parallel Coordinates** (with scaling and brushing/highlighting).
 - **Projection technique:** at least one among PCA, t-SNE, MDS, LAMP, or UMAP.

Additional supporting plots (scatterplots, histograms, correlation matrices, etc.) are encouraged.

Evaluation Criteria

This assignment is worth 20 points. I will evaluate both the strength of your design and the quality of your computing process. I will also consider factors such as the target audience, the clarity of your message, and how well the intended task is addressed. Your work will be grade based on:

- **Clarity:** Is the visualization easy to interpret?
- **Analytical Insight:** Does it reveal interesting insights?
- **Design Rationale:** Are your choices well-motivated?
- **Technical Accuracy:** are techniques used adequately?
- **Tasks:** The visualization help to answer 3 tasks effectively.
- **Interaction:** could you alpply some interactionsand animation resources such as: mouseenter, mouseover, mouseclick, and so on?

Deliverables instructions

- Your coded solutions must use D3. Other visualization libraries are not allowed.
- Submit to canvas a single ZIP containing a flask application and data files (nodes and edges).
- You are required to present your work in class. Failure to present will result in no grade for this assignment.

**The IN-PERSON presentation in
class is mandatory**